Integrating Regularization and PnP Priors for SAR Image Reconstruction Using Multiagent Consensus Equilibrium

Yizhe Fan, Bingchen Zhang, and Yirong Wu

***Abstract*— The multiagent consensus equilibrium (MACE) mechanism, which generalizes the popular method plug-and-play (PnP)-alternating direction method of multipliers (ADMM) and composite regularization in computational sensing, possesses the notable capacity to incorporate multiple priors aspect of both regularization and PnP for improving image quality. In this work, a flexible synthetic aperture radar (SAR) image reconstruction method based on MACE is proposed to integrate multiple regularization and PnP priors for various features enhancement. The partial-update approach and Mann iteration methods are implemented to increase the computational efficiency of the MACE-based SAR image reconstruction algorithm. A thorough analysis of the proposed algorithm's convergence and computational complexity is provided. High-quality SAR images necessitate low ambiguity, high target-to-background ratio (TBR), and low coherent speckle. We therefore demonstratively integrate regularization and PnP priors for azimuth ambiguity suppression, sparsity inducing, multiple features enhancement, and despeckling. The proposed method's performance is evaluated through experiments on both simulated and QILU-1 satellite SAR data.**

***Index Terms*— Multiagent consensus equilibrium (MACE), SAR image reconstruction, sparse signal processing, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is the major technology of modern microwave imaging remote sensing and has the all-time and all-weather observing ability. The combination of sparse signal processing and SAR image formation can potentially improve the performance and reduce system data complexity. Prior knowledge of analytical regularization terms and plug-and-play (PnP) denoisers is introduced to sparse microwave imaging for denoising and various features enhancement.

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Initially, ℓ_1-norm is introduced as prior for sparse reconstruction of SAR image [1]. In recent years, composite regularization models with multiple analytical priors are utilized to integrate image enhancement with formation. For instance, combining the ℓ_1 and total variation (TV) norms to enhance region-based features while imaging [2], [3], and the ℓ_1-norm and group sparsity constraint [4] to suppress azimuth ambiguity and noise [5]. In the pursuit of more accurate reconstruction, nonconvex penalties are introduced as an alternative to the ℓ_1-norm as the sparsity-inducing penalty term [6], [7]. Alternating direction method of multipliers (ADMM) is one of the most widely used and efficient algorithms for solving composite regularization problems. However, it remains a formidable challenge to integrate nonanalytical priors, such as deep learning operators and denoising filters, within regularization. ADMM splits the primal problem into a sequence of subproblems through variable splitting. Furthermore, finding solutions to subproblems can be extremely time-consuming when complicated priors are used.

To overcome the limitation of analytical prior knowledge, the PnP concept is adopted, where the update step of the ADMM is replaced with a denoiser prior, leading to the conceptualization of the PnP-ADMM method [8]. Subsequently, methods such as PnP-ISTA and PnP-FISTA are developed to accommodate additional feature enhancement goals [9]. The ability of PnP methods to incorporate pretrained data fidelity priors into sparse signal processing has made them a widely adopted framework for computational imaging, including SAR image reconstruction [10], [11], [12]. Except for the enhancement in image quality, the implementation of PnP priors substantially reduces the computational complexity associated with solving subproblems of ADMM. The regularization by denoising (RED) [13], [14] method provides a distinct way to develop the PnP concept. Instead of using denoisers as priors in the PnP-ADMM method, RED incorporates denoisers and regularization models through formulating the denoiser as a regularization term in the objective function. Nonetheless, the superiority of imaging performance for RED compared to PnP-ADMM has not been sufficiently demonstrated with sufficient evidence [15].

Multiagent consensus equilibrium (MACE), generalized from PnP-ADMM, is capable of accommodating multiple priors, which represent different objectives [16]. MACE extends the composite regularization to allow for both regularization and PnP priors, offering flexibility in the selection of objec-

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tives, operators, as well as algorithms. MACE has shown its potential to integrate regularization penalties and deep learning priors with imaging models in various fields of computational imaging, including computerized tomography (CT) reconstruction [17], [18], [19], [20], magnetic resonance imaging (MRI) [19], ptychotomography [21], quanta image sensors (QIS) [22], microscopy [23], seismic shots recovery [24], and ultrasound imaging [25]. Numerous papers have addressed the convergence of the PnP-ADMM method, with some providing local convergence under specific limitation of the denoiser [26], [27]. Nevertheless, the convergence criteria of nonanalytical implicit priors in SAR imaging have not been mathematically justified.

In this article, we establish a novel SAR image reconstruction method based on MACE mechanism. The proposed method is adaptable to a wide range of regularization and PnP priors. High-quality SAR images necessitate low ambiguity, high target-to-background ratio (TBR), and low coherent speckle. To cope with these issues, we integrate a comprehensive set of priors for instance, including group sparsity constraint for azimuth ambiguity suppression [4], nonconvex penalty [6] and dictionary-based regularization [28], [29] for feature enhancement, and TV-norm and BM3D [30] for despeckling. BM3D is a PnP prior while the rest are regularization priors. In brief, the MACE mechanism can be used to improve reconstruction quality by fusing multiple sources of information implemented as priors such as regularization penalties, pretrained networks, and denoising filters. It is significant to note that the MACE agents are not restricted to only nonconvex penalty, TV-norm, or BM3D denoiser. Instead, our proposed method can integrate a wide range of other prior models, such as deep learning networks for ambiguity suppression [31] and CNN-based denoisers [10], to improve imaging performance.

The main contributions of this article are as follows.

1) A flexible SAR image reconstruction method based on MACE is established to integrate multiple regularization and PnP priors for various features enhancement.
2) The partial-update approach and Mann iteration are introduced to improve the computational efficiency of MACE-based SAR image reconstruction. The convergence and computational complexity of the proposed algorithm is analyzed.
3) The proposed method integrates azimuth ambiguity suppression, feature enhancement, and despeckling agents to obtain high-quality reconstructed SAR images. The performance is evaluated via simulated experiments and real data of QILU-1 SAR satellite.

The remainder of this article is organized as follows. The general idea of MACE mechanism and MACE-based SAR reconstruction method is explained in Section II. The agents design for the proposed method is analyzed in detail in Section III. Section IV illustrates the details of calculating the solution to the proposed method, as well as the analysis of convergence and computational complexity. In Section V, experimental results and analysis of MACE-based SAR reconstruction method's application are presented. Finally, the conclusion is drawn in Section VI.

II. SAR Image Reconstruction Model Based on MACE Mechanism

This section combines the MACE mechanism [16] with sparse microwave imaging model, and establishes a high-quality SAR image reconstruction method.

A. Sparse SAR Imaging Model

The sparse microwave imaging model for SAR image reconstruction can be expressed as follows [1]:

$$\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{x} + \boldsymbol{n} \tag{1}$$

where $\boldsymbol{y} \in \mathbb{C}^{M\times 1}$ denotes the vector form of echo, $\boldsymbol{x} \in \mathbb{C}^{N\times 1}$ denotes the vector form of image, $\boldsymbol{\Phi} \in \mathbb{C}^{M\times N}$ denotes the measurement matrix of the imaging system, and $\boldsymbol{n} \in \mathbb{C}^{M\times 1}$ denotes the additive Gaussian white noise vector.

The measurement matrix of a typical stripmap SAR can be expressed as follows [1]:

$$\boldsymbol{\Phi} = \begin{bmatrix} \phi(1,1) & \phi(1,2) & \dots & \phi(1,N) \\ \phi(2,1) & \phi(2,2) & \dots & \phi(2,N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(M,1) & \phi(M,2) & \dots & \phi(M,N) \end{bmatrix} \tag{2}$$

where $\phi(m,n)$ denotes the element of measurement matrix at the mth row and nth column, written as follows:

$$\phi(m,n) = \int\int_{(t,\tau)\in T_m} \omega_a\left(t - \frac{p_n}{\mathrm{v}}\right) \cdot s\left(\tau - \frac{2R(p_n,q_n,t)}{c}\right) \cdot \exp\left\{-j4\pi f_0 \frac{R(p_n,q_n,t)}{c}\right\} \mathrm{d}t\,\mathrm{d}\tau \tag{3}$$

where (t,τ) denote the azimuth time and range time, respectively, T_m represents the mth time series, $\omega_a(\cdot)$ is the azimuth weighting function, v is the equivalent platform velocity relative to the ground, $s(\cdot)$ is the transmitting signal, $R(p_n,q_n,t)$ represents the slant range between the target located at position (p_n,q_n) and the radar at time t, c is the speed of light, and f_0 is the carrier frequency.

B. MACE Mechanism

The general idea of MACE-based SAR reconstruction is illustrated in Fig. 1. Consensus without equilibrium implies that distinct agents reach identical outputs, whereas equilibrium without consensus signifies that the sum of each agent's update vector equates to zero. Accordingly, the MACE mechanism integrates multiple agents by offsetting the effects of each agent $\boldsymbol{v}_i^* = \boldsymbol{v}_i - \hat{\boldsymbol{v}}$.

A MACE agent $F_i : \mathbb{C}^N \to \mathbb{C}^N$ is a function that takes the input image $\boldsymbol{v}_i$ and produces an improved output image. In CT reconstruction, for instance, a denoiser and proximal maps are examples of agents, but other agents might implement artifact removal deep learning networks and heuristic improvements [15]. The idea of achieving the consensus equilibrium

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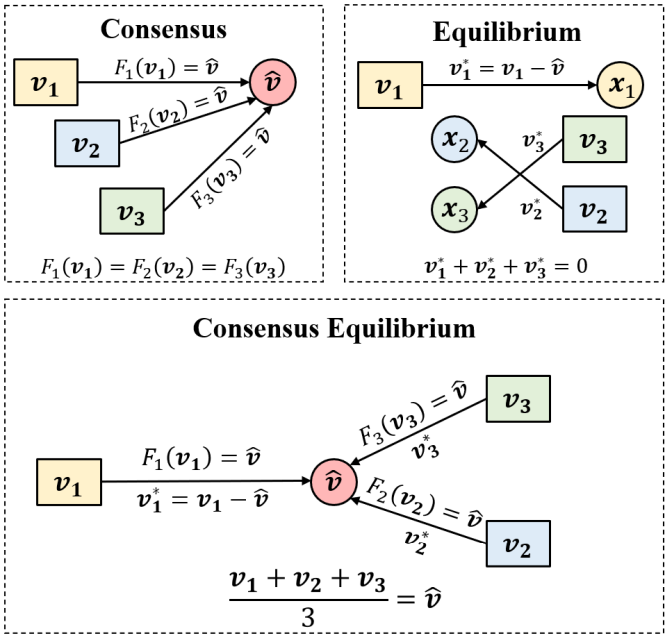


Fig. 1. Geometric implementation of MACE mechanism. Consensus implies that distinct agents reach identical outputs. Equilibrium signifies that the sum of each agent's update vector equates to zero.

among multiple agents can be formulated as solving the consensus equilibrium equations [16]

$$F_i\left(\hat{\boldsymbol{v}} + \boldsymbol{v}_i^*\right) = \hat{\boldsymbol{v}} \quad (i = 1, 2, \ldots, L), \qquad \sum_{i=1}^{L} \boldsymbol{v}_i^* = 0 \tag{4}$$

where F_i denotes the ith agent, $\hat{\boldsymbol{v}}$ denotes the consensus equilibrium solution, which is the imaging result, $\boldsymbol{v}_i^*$ denotes the force applied by the ith agent that balances to zero, and L is the number of agents.

The MACE solution is given by the consensus equilibrium equations that arise simultaneously from the balance of multiple agents, which may include various kinds of image processing operations. In the case of regularization inversion models, for which regularization priors are introduced to build objective function, the corresponding MACE agents can be formulated as follows:

$$F_i(\boldsymbol{v}) = \arg\min_{\boldsymbol{x}} f_i(\boldsymbol{x}) + \eta \|\boldsymbol{x} - \boldsymbol{v}\|_2^2 \tag{5}$$

where $\boldsymbol{v}$ denotes the vector form of the input image, $\|\cdot\|_2$ denotes the ℓ_2-norm, and η is a user-selected parameter that will ultimately affect the convergence speed of the algorithm. The objective function f_i denotes the regularization priors used for generating the agent. Intuitively, the agent F_i takes an input $\boldsymbol{v}$, and returns an image that reduces its associates objective function and is close to $\boldsymbol{v}$.

Any regularization prior can be utilized as an agent by (5), but MACE also applies to a wide range of operations, which we regard as the PnP priors, for which there is no corresponding objective functions. In this case, the PnP priors are compatible by

$$F_j(\boldsymbol{v}) = \text{PnP}(\boldsymbol{v}) \tag{6}$$

where $\text{PnP}(\cdot)$ denotes the PnP priors for image-domain operation.

In Section III, we will demonstrate in detail the process of generating agents from regularization penalties or PnP priors for SAR image quality enhancement.

C. MACE-Inspired SAR Image Reconstruction

The optimization framework of sparse microwave imaging is generally written as a composite regularization problem

$$\hat{\boldsymbol{x}} = \arg\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{x}\|_2^2 + \sum_{i=1}^{L'} \lambda_i p_i(\boldsymbol{x}) \tag{7}$$

where $\hat{\boldsymbol{x}}$ denotes the imaging result, $\|\cdot\|_2$ denotes the ℓ_2-norm, λ_i denote the regularization parameters controlling penalty terms $p_i(\cdot)$, and L' denotes the number of penalty terms. Penalty terms (p_i), also known as regularization priors, are customized based on specific demands, such as ℓ_1-norm, TV-norm [2], nonconvex penalty [6], and $\ell_{2,1}$-norm [5].

Inspired by the MACE mechanism, we relate the composite regularization model (7) to MACE agents (5) by letting

$$F_1(\boldsymbol{v}) = \arg\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{x}\|_2^2 + \lambda p_1(\boldsymbol{x}) + \frac{\eta}{2}\|\boldsymbol{x} - \boldsymbol{v}\|_2^2 \tag{8}$$

$$F_i(\boldsymbol{v}) = \arg\min_{\boldsymbol{x}} p_i(\boldsymbol{x}) + \frac{\eta}{2}\|\boldsymbol{x} - \boldsymbol{v}\|_2^2, i = 2, 3, \ldots, L' \tag{9}$$

where $\boldsymbol{v} \in \mathbb{C}^{N\times 1}$ denotes the vector form of the input complex-valued image, λ denotes the regularization parameter controlling sparsity-inducing penalty term $p_1(\cdot)$, and η is trivially set to 1. F_1 is a sparse reconstruction agent with prior p_1, which can be any sparsity-inducing priors, such as ℓ_1-, ℓ_q-, or $\ell_{2,1}$-norm. The other F_i utilizes image-domain priors for improving reconstruction quality. By reformulating the residual term $\|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{x}\|_2^2$ and penalty term p_i as individual agents F_i, the consensus equilibrium equations can achieve exactly the same solution of (7), $\hat{\boldsymbol{v}} = \hat{\boldsymbol{x}}$ [16]. With this approach, the high computational complexity associated with solving composite regularization problems is eliminated.

The reconstruction agent F_1 has to store and process the large-scale measurement matrix, which would bring huge memory and computational cost. Thus, we adopt the azimuth-range decouple operators [5] in our method. Azimuth-range decouple operators can be implemented to substitute the measurement matrix and its Hermitian transpose [32]. Supposing that $\mathcal{G}(\cdot)$ is the raw data generation operator and $\mathcal{I}(\cdot)$ is the imaging operator, the azimuth-range decouple scheme is formulated as

$$\mathcal{G}(\boldsymbol{x}) \cong \boldsymbol{\Phi}\boldsymbol{x}, \quad \mathcal{I}(\boldsymbol{x}) \cong \boldsymbol{\Phi}^H\boldsymbol{x}. \tag{10}$$

Applying these two operators to physics-based agents can effectively reduce computational complexity [6]. By paralleling the calculation of each agents during iteration, the results of the proposed method can be acquired efficiently.

Furthermore, MACE's notable advantage is that we can not only generate agents from regularization models but utilize PnP priors with no analytical representation as well, such as BM3D denoisers [30] or deep learning networks of image enhancement

$$F_j(\boldsymbol{v}) = \text{PnP-SAR}(\boldsymbol{v}), \quad j = \left(L' + 1\right), \ldots, L \tag{11}$$

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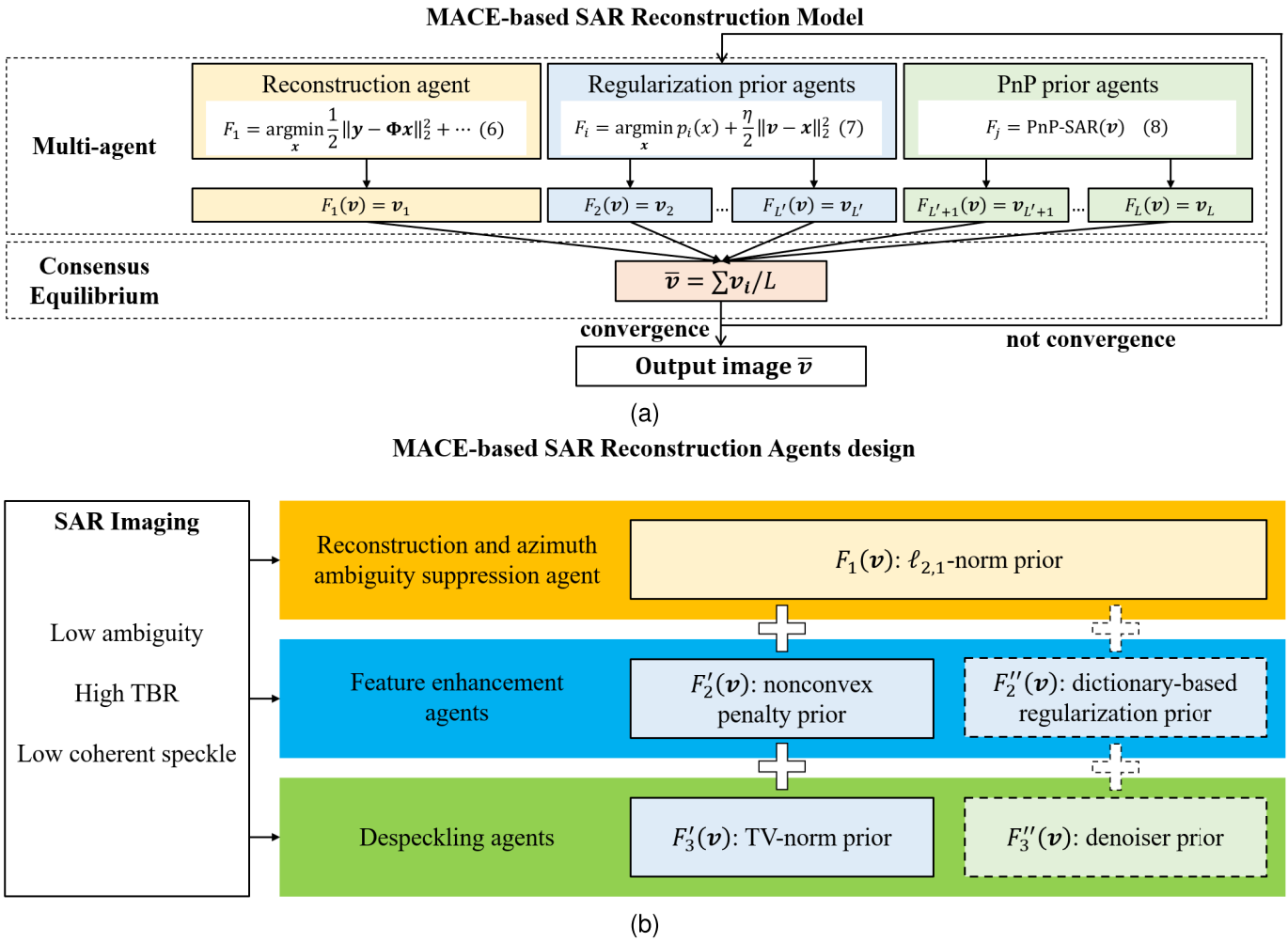


Fig. 2. MACE-based SAR reconstruction model. The proposed method employs different sets of agents, categorized in (a) based on the type of prior that they utilize, while the agents in (b) are categorized based on their effect on improving reconstruction quality.

where PnP-SAR$(\cdot)$ denotes the PnP priors for SAR image processing, and L is the total amount of agents.

Hence, for both regularization and PnP priors, MACE encompasses the original composite regularization method but also allows for a wider variety of operators and solution algorithms. Fig. 2(a) illustrates the general structure of our proposed method.

III. Agents Design for SAR Image Reconstruction

Regularization priors and PnP priors play distinct roles in certain SAR image formation requirements and enhancements. Our proposed method therefore incorporates both these types of priors, leveraging their individual advantages via MACE. Specifically, with the goal of reconstructing low ambiguity, high TBR, and low-coherent speckle SAR image, three sets of agents, which are illustrated in Fig. 2(b), are utilized in the proposed method, including an azimuth ambiguity-free reconstruction agent based on $\ell_{2,1}$ prior, feature enhancement agents based on nonconvex penalty or dictionary-based regularization, and despeckling agents using TV-norm or BM3D denoiser. In contrast to Fig. 2(a), where the agents are classified based on the type of priors they utilize, Fig. 2(b) classifies agents in the proposed method based on their effect on improving reconstruction quality. We demonstrate the capabilities of the MACE-based SAR imaging method by exploring various priors for a single purpose, showing that the method can effectively incorporate a diverse range of priors. Detailed account of each agent is given in the following.

A. Reconstruction and Azimuth Ambiguity Suppression Agent F_1

We introduce the group sparsity [4], [5] penalty for azimuth ambiguity suppression while reconstruction. The group sparsity constraint is imposed in the sparse SAR imaging model to tackle azimuth ambiguities due to the fact that the main imaging area ($\boldsymbol{x}_0$) and ambiguous areas ($\boldsymbol{x}_{\pm 1}$) share the same signal support set. Taking the elements at the same position in $\boldsymbol{x}_j (j = 0, \pm 1)$ as a group, their value should be zero or nonzero at the same time.

Agent F_1 is generated from a regularization prior in form (8), that is, to minimize the function

$$F_1(\boldsymbol{v}) = \arg\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{\Phi x}\|_2^2 + \lambda\|\boldsymbol{x}\|_{2,1} + \frac{\eta}{2}\|\boldsymbol{x} - \boldsymbol{v}\|_2^2 \quad (12)$$

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where $\|\cdot\|_{2,1}$ denotes the $\ell_{2,1}$-norm for group sparsity penalty, which can be written as follows:

$$\|\boldsymbol{x}\|_{2,1} = \sum_{i=1}^{N} \sqrt{\sum_{j=-1,0,1} |x_{j,i}|^2} \tag{13}$$

where operator $|\cdot|$ represents the magnitude calculation, and $\boldsymbol{x}_j = \{x_{j,i}\}(j = -1, 0, 1)(i = 1, 2, \ldots, N)$ denote the backscattering coefficient of the main imaging area and the ± 1 ambiguities areas, respectively.

In agent F_1, the objective function consists of a residual term and a group sparse penalty term. Group sparse penalty can effectively suppress azimuth ambiguities to improve image reconstruction quality. In addition, the $\ell_{2,1}$-norm also plays the role of sparsity-inducing penalty in our method.

B. Feature Enhancement Agents F_2

The existing feature enhancement priors include ℓ_1-norm, nonconvex (NC) [6] penalty such as minimax concave (MC) penalty and ℓ_q-norm for point-based feature enhancement, and dictionary-based reconstruction [28] for target-oriented enhancement. In this article, both MC penalty and dictionaries are utilized for multiple feature enhancement

$$F_2'(\boldsymbol{v}) = \arg\min_{\boldsymbol{x}} \mathrm{MC}(\boldsymbol{x}) + \frac{\eta}{2}\|\boldsymbol{x} - \boldsymbol{v}\|_2^2 \tag{14}$$

where $\mathrm{MC}(\cdot)$ denotes the minimax concave penalty

$$\mathrm{MC}(\boldsymbol{x}) = \sum_{k=1}^{N} \begin{cases} |x_i| - \dfrac{|x_i|^2}{2\theta}, & |x_i| \le \theta \\ \theta/2, & |x_i| > \theta \end{cases} \tag{15}$$

where $\theta > 1$ is the threshold of MC penalty, and the operator $|\cdot|$ represents the magnitude calculation. Since the MC penalty is closer to ℓ_0-norm compared with ℓ_1-norm, the amplitude error of point targets caused by bias effect could be alleviated [6]. With MC penalty utilized as prior, F_2' simultaneously performs the functions of a point-based feature enhancement agent and ensures the accuracy of the reconstruction amplitude.

Another feature enhancement agent is generated from dictionary-based regularization prior

$$F_2''(\boldsymbol{v}) = \boldsymbol{\Theta}\boldsymbol{D} \arg\min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_p^p + \frac{\eta}{2}\|\boldsymbol{\Theta}\boldsymbol{D}\boldsymbol{\alpha} - \boldsymbol{v}\|_2^2 \tag{16}$$

where $\boldsymbol{\Theta} = \mathrm{diag}\{v_i/|v_i|\}$ denotes the diagonal matrix of the complex-value vector $\boldsymbol{v}$'s imaginary part, $\boldsymbol{D}$ is an overcomplete dictionary, and $\boldsymbol{\alpha}$ is the vector of representation coefficients, satisfying $\boldsymbol{x} = \boldsymbol{\Theta}\boldsymbol{D}\boldsymbol{\alpha}$. The performance of feature enhancement hinges heavily on the selection of dictionary, therefore it is essential to design a dictionary $\boldsymbol{D}$ that aligns with the application and targets of interest. This dictionary aims to provide a sparse representation of the magnitude of complex-valued image which contains the features of interest within the scene, such as overcomplete shape-based dictionary for synthetic or simple real scenes, wavelet dictionary for man-made targets with stronger reflectivities [33], or learned target-oriented dictionaries [34].

F_1 plays the role of the sparsity-inducing operator, while F_2' and F_2'' are agents for multiple feature enhancement, including the point-based and target-oriented features. Now the task at hand is to design an agent that can enhance region-based features, regarding the speckles effect in SAR images. To cope with these issues, in Section III-C, a PnP prior and a regularization prior are employed for despeckling.

C. Despeckling Agents F_3

BM3D [30] has been widely adopted within the PnP framework for SAR reconstruction image despeckling [10], [12]. The agent generated from PnP priors can be formulated as (6)

$$F_3'(\boldsymbol{v}) = \mathrm{BM3D}(\boldsymbol{v}) \tag{17}$$

where $\mathrm{BM3D}(\cdot)$ denotes the BM3D denoiser, which is the PnP prior utilized in our method.

TV-norm, which is a classic regularization penalty, is another widely used prior for despeckling [35]. Here we introduce TV-norm as an alternative agent for despeckling to show the flexibility of the proposed MACE-based reconstruction

$$F_3''(\boldsymbol{v}) = \mathrm{TV}(|\boldsymbol{x}|) + \frac{\eta}{2}\|\boldsymbol{x} - \boldsymbol{v}\|_2^2. \tag{18}$$

The TV-norm $\mathrm{TV}(\cdot)$ of image vector $\boldsymbol{x}$ is defined as follows:

$$\mathrm{TV}(\boldsymbol{x}) = \sum_{i,j} \sqrt{\left[\nabla_h |\boldsymbol{X}|_{i,j}\right]^2 + \left[\nabla_v |\boldsymbol{X}|_{i,j}\right]^2} \tag{19}$$

where $\boldsymbol{X}$ denotes the 2-D complex-valued matrix corresponding to image vector $\boldsymbol{x}$, $|\boldsymbol{X}|$ denotes the elementwise magnitude calculation, and $\nabla_h|\boldsymbol{X}|_{i,j}$ and $\nabla_v|\boldsymbol{X}|_{i,j}$ denote the gradient of pixel in the ith row and jth column, which is defined as follows:

$$\begin{aligned} \nabla_h|\boldsymbol{X}|_{i,j} &= |\boldsymbol{X}|_{i+1,j} - |\boldsymbol{X}|_{i,j} \\ \nabla_v|\boldsymbol{X}|_{i,j} &= |\boldsymbol{X}|_{i,j+1} - |\boldsymbol{X}|_{i,j}. \end{aligned} \tag{20}$$

Two kinds of despeckling priors are introduced in this article, the TV-norm and BM3D operator, both of which are feasible in MACE-based SAR reconstruction method.

It is crucial to acknowledge that MACE agent F_3 is not limited to TV-norm, or BM3D denoiser. Instead, our proposed method can integrate a wide range of other prior models, such as deep learning network-based denoisers [10], [36]. Our method can also incorporate priors for azimuth ambiguity suppression [31] or targets feature enhancement [37] as F_1 or F_2 to improve imaging performance.

IV. Algorithm of the Proposed SAR Image Reconstruction Method

We convert the consensus equilibrium equations as a fixed-point problem and derive the step-by-step solution to the proposed SAR image reconstruction MACE model. The convergence and computational complexity are discussed in this section.

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A. Solution to MACE Equations

In order to represent the MACE equations of (4) in a more compact form, we first define a stacked matrix

$$\boldsymbol{V} = \left[\boldsymbol{v}_1\ \boldsymbol{v}_2\ \ldots\ \boldsymbol{v}_L.\right] \in \mathbb{C}^{NL\times 1}. \tag{21}$$

Then we define the stacked multiagent operator $\boldsymbol{F}$

$$\boldsymbol{F}(\boldsymbol{V}) = \left[F_1(\boldsymbol{v}_1)\ F_2(\boldsymbol{v}_2)\ \cdots\ F_L(\boldsymbol{v}_L)\right] \in \mathbb{C}^{NL\times 1} \tag{22}$$

where F_i denotes the ith agent in the framework. Next we generate the averaging operator $\boldsymbol{G_u}$

$$\boldsymbol{G_u}(\boldsymbol{V}) = \left[\bar{\boldsymbol{v}}\ \bar{\boldsymbol{v}}\ \cdots\ \bar{\boldsymbol{v}}\right](L \text{ items}) \in \mathbb{C}^{NL\times 1} \tag{23}$$

where $\boldsymbol{u} = [u_1, u_2, \ldots, u_L]$ is the weighting coefficient and $\bar{\boldsymbol{v}}$ denotes the weighted mean of $\boldsymbol{V}$'s column vectors

$$\bar{\boldsymbol{v}} = \frac{\sum_{i=1}^{L} u_i \boldsymbol{v}_i}{\sum_{i=1}^{L} u_i}. \tag{24}$$

Theorem 1: The MACE equations of (4) have the much more compact form of

$$\boldsymbol{F}(\boldsymbol{V}) = \boldsymbol{G_u}(\boldsymbol{V}) \tag{25}$$

with the solution given by $\hat{\boldsymbol{V}} = \left[\hat{\boldsymbol{v}}_1\ \hat{\boldsymbol{v}}_2\ \ldots\ \hat{\boldsymbol{v}}_L\right]$, where $\hat{\boldsymbol{v}}_i = \hat{\boldsymbol{x}} + \boldsymbol{x}_i^*$. $\hat{\boldsymbol{x}}$ denotes the imaging results and satisfies $\hat{\boldsymbol{x}} = \bar{\hat{\boldsymbol{v}}}$. $\bar{\hat{\boldsymbol{v}}}$ is calculated by (24).

Proof: Let $(\hat{\boldsymbol{x}}, \boldsymbol{x}_i^*)$ be a solution to (4), then $\hat{\boldsymbol{v}}_i = \hat{\boldsymbol{x}} + \boldsymbol{x}_i^*$ satisfies $F_i(\hat{\boldsymbol{v}}_i) = F_i(\hat{\boldsymbol{x}} + \boldsymbol{x}_i^*) = \hat{\boldsymbol{x}}$. Let $\bar{\hat{\boldsymbol{v}}}$ denote the weighted mean entries of $\boldsymbol{G_u}(\hat{\boldsymbol{V}})$, which can be computed using (24). Since $\sum_{i=1}^{L} \boldsymbol{x}_i^* = 0$, we have

$$\bar{\hat{\boldsymbol{v}}} = \frac{\sum_{i=1}^{L} u_i \hat{\boldsymbol{v}}_i}{\sum_{i=1}^{L} u_i} = \frac{\sum_{i=1}^{L} u_i \left(\hat{\boldsymbol{x}} + \boldsymbol{x}_i^*\right)}{\sum_{i=1}^{L} u_i} = \hat{\boldsymbol{x}}. \tag{26}$$

Thus, $\boldsymbol{F}(\hat{\boldsymbol{V}}) = \boldsymbol{G_u}(\hat{\boldsymbol{V}})$. Conversely, let $\hat{\boldsymbol{V}}$ be a solution to (25). Equation (4) can be satisfied by letting $\hat{\boldsymbol{x}} = \bar{\hat{\boldsymbol{v}}}$ and $\boldsymbol{x}_i^* = \hat{\boldsymbol{v}}_i - \hat{\boldsymbol{x}}$. □

The significance of Theorem 1 and its proof is the elucidation of the interconnection between imaging results and solutions to MACE problems. While operating the imaging results $\hat{\boldsymbol{x}}$ with an individual agent F_i, there will be a bias $\boldsymbol{x}_i^*$ between the output $\hat{\boldsymbol{v}}_i$ and the imaging result. The function of the MACE mechanism is to balance all biases to zero and achieve a consensus result.

Then we solve the problem based on the following theorem, which is discussed in detail and proved in [16], Corollary 3.

Theorem 2: Let $\boldsymbol{T} = (2\boldsymbol{G_u} - \boldsymbol{I})(2\boldsymbol{F} - \boldsymbol{I})$. $\hat{\boldsymbol{V}}$ is a solution to (25) if and only if

$$\boldsymbol{T}\hat{\boldsymbol{V}} = \hat{\boldsymbol{V}}. \tag{27}$$

Theorems 1 and 2 convert the problem to a fixed-point problem, which can be efficiently solved by the well-known Mann iteration method [38]. The fixed-point solution of (27) can be acquired by iteration with a step size parameter $\rho \in (0, 1)$

$$\boldsymbol{V}^{k+1} = (1-\rho)\boldsymbol{V}^k + \rho \boldsymbol{T}\boldsymbol{V}^k. \tag{28}$$

From (28), we have

$$\begin{aligned}\boldsymbol{V}^{k+1} &= \boldsymbol{V}^k + \rho(\boldsymbol{T} - \boldsymbol{I})\boldsymbol{V}^k \\ &= \boldsymbol{V}^k + \rho[(2\boldsymbol{G_u} - \boldsymbol{I})(2\boldsymbol{F} - \boldsymbol{I}) - \boldsymbol{I}]\boldsymbol{V}^k \\ &= \boldsymbol{V}^k + \rho[2\boldsymbol{G_u}(2\boldsymbol{F} - \boldsymbol{I}) - 2\boldsymbol{F}]\boldsymbol{V}^k \\ &= \boldsymbol{V}^k + 2\rho\left[\boldsymbol{G_u}\left(2\boldsymbol{F}\left(\boldsymbol{V}^k\right) - \boldsymbol{V}^k\right) - \boldsymbol{F}\left(\boldsymbol{V}^k\right)\right].\end{aligned} \tag{29}$$

Let $\boldsymbol{Z}^k = \boldsymbol{F}(\boldsymbol{V}^k)$ and $\boldsymbol{U}^k = \boldsymbol{G_u}(2\boldsymbol{Z}^k - \boldsymbol{V}^k)$, the pseudo-code for MACE-SAR method is provided in Algorithm 1.

Algorithm 1 General MACE-Based SAR Image Reconstruction

Input: $\boldsymbol{\Phi}$, $\boldsymbol{y}$, $\boldsymbol{V} = \boldsymbol{V}^0$, $\boldsymbol{u}$, ρ

Generate reconstruction agent.
Generate regularization-prior agents.
Generate PnP-prior agents.
while not converged **do**
  $\boldsymbol{Z} = \boldsymbol{F}(\boldsymbol{V})$
  $\boldsymbol{U} = \boldsymbol{G_u}(2\boldsymbol{Z} - \boldsymbol{V})$
  $\boldsymbol{V} = \boldsymbol{V} + 2\rho(\boldsymbol{U} - \boldsymbol{Z})$
end while

Output: $\hat{\boldsymbol{x}} = \bar{\hat{\boldsymbol{v}}}$

B. Convergence Analysis

While the guaranteed convergence of the Mann iteration for the map $\boldsymbol{T}$ with Lipschitz constant $L(\boldsymbol{T}) \le 1$ is provided in [39], the use of SAR observation matrix and priors in our proposed method may not satisfy this condition. Consequently, the convergence of our method requires further analysis.

In the case of $L = 2$, $\rho = 0.5$, $\boldsymbol{u} = (0.5, 0.5)$, and all the agents are generated by regularization prior, the iteration is equivalent to the classic variable-splitting ADMM algorithm

$$\begin{aligned}\begin{bmatrix}\boldsymbol{v}_1^{k+1} \\ \boldsymbol{v}_2^{k+1}\end{bmatrix} &= \frac{1}{2}\begin{bmatrix}\boldsymbol{v}_1^k \\ \boldsymbol{v}_2^k\end{bmatrix} + \frac{1}{2}(2\boldsymbol{G_u} - \boldsymbol{I})\begin{bmatrix}2F_1\left(\boldsymbol{v}_1^k\right) - \boldsymbol{v}_1^k \\ 2F_2\left(\boldsymbol{v}_2^k\right) - \boldsymbol{v}_2^k\end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix}\boldsymbol{v}_1^k + 2F_2\left(\boldsymbol{v}_2^k\right) - \boldsymbol{v}_2^k \\ \boldsymbol{v}_2^k + 2F_1\left(\boldsymbol{v}_1^k\right) - \boldsymbol{v}_1^k\end{bmatrix}.\end{aligned} \tag{30}$$

Hence, the ADMM-based algorithms for composite regularization SAR reconstruction models [5], [6] ensure the convergence of the proposed method for regularization priors with $\rho = 0.5$. In some cases, $\rho > 0.5$ leads to faster convergence speed. Bauschke and Combettes [40] replace the scalar ρ with a sequence ρ_k that obeys $\sum \rho_k(1 - \rho_k) = \infty$ to promote the convergence of iteration. Buzzard et al. [16] suggest using an alternative preconditioning symmetric positive definite matrix $\boldsymbol{H}$ instead of ρ that may increase the convergence speed.

In fact, the iteration in (28) can still converge for more general $\boldsymbol{T}$. Suppose $\boldsymbol{T}$ is differentiable at a fixed point $\boldsymbol{W}$, then the Jacobian matrix of $\boldsymbol{T}$ at $\boldsymbol{W}$ gives the linearization of $\boldsymbol{T}$. Since the fixed point $\boldsymbol{W}$ satisfies (27), iteration (28) maintains the fixed point by (suppose $\boldsymbol{V}^k = \boldsymbol{W}$)

$$(1-\rho)\boldsymbol{W} + \rho\boldsymbol{T}\boldsymbol{W} = (1-\rho)\boldsymbol{W} + \rho\boldsymbol{W} = \boldsymbol{W}. \tag{31}$$

Let $\boldsymbol{A}$ denote the Jacobian matrix of $\boldsymbol{T}$ at $\boldsymbol{W}$, and $\boldsymbol{e}$ denote the eigenvalues of $\boldsymbol{A}$. Then, we have $\boldsymbol{T}\boldsymbol{V}^k \approx \boldsymbol{W} + \boldsymbol{A}(\boldsymbol{V}^k - \boldsymbol{W})$, where $\boldsymbol{V}^k$ is the input of the kth iteration and is in the neighborhood of fixed point $\boldsymbol{W}$. The iteration can be approximated

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as follows:

$$\begin{aligned} \boldsymbol{V}^{k+1} - \boldsymbol{W} &= (1-\rho)\boldsymbol{V}^k + \rho \boldsymbol{T}\boldsymbol{V}^k \\ &\approx (1-\rho)\boldsymbol{V}^k + \rho\big[\boldsymbol{W} + \boldsymbol{A}\big(\boldsymbol{V}^k - \boldsymbol{W}\big)\big] - \boldsymbol{W} \\ &= [(1-\rho) + \rho\boldsymbol{A}]\big(\boldsymbol{V}^k - \boldsymbol{W}\big). \end{aligned} \tag{32}$$

Therefore, the eigenvalues of $[(1-\rho)+\rho\boldsymbol{A}]$ are given by $\boldsymbol{g} = (1-\rho) + \rho\boldsymbol{e}$.

When the real part of each eigenvalue e_i is less than 1, we can shrink the eigenvalues g_i to lie strictly inside the unit disk of the complex plane by taking sufficiently small ρ. Then come to the following theorem.

Theorem 3: Suppose that $\boldsymbol{T}$ has a fixed point $\boldsymbol{W}$ and is differentiable at $\boldsymbol{W}$. Let $e_1, \ldots, e_{NL}$ denote the eigenvalues of the Jacobian matrix of $\boldsymbol{T}$ at $\boldsymbol{W}$. For all i, the real part of e_i is strictly less than 1, then there exist $\rho \in (0, 1)$ and an open set Ω containing $\boldsymbol{W}$ such that for any initial point $\boldsymbol{V}^0$ in Ω, the iteration defined by (28) converges to $\boldsymbol{W}$.

Proof: See Appendix for detailed proof. □

For reconstruction and regularization prior agents, the corresponding real parts of eigenvalues can be written as $\eta/(1+\eta)$, which is strictly less than 1. However, there is no constraint to eigenvalues when PnP priors are utilized. In this case, the convergence can be optimized by limiting the eigenvalues in a disk of minimum radius. While there is no guarantee of Lipschitz constant for any general PnP − SAR(·), in practice, we have found that the Mann iteration converges. This is consistent with previous experimental results that have empirically observed convergence of PnP-based SAR imaging for a wide variety of denoisers including BM3D [11], [12], or deep CNNs [10].

C. Partial Update of Agent Operators

Due to the iterative nature of the regularization-prior agents F_1, F_2', F_2'', and F_3'', a direct implementation of the MACE iteration (28) would be impractical. This results in multiple iterations of complex optimization, which ultimately leads to a computationally inefficient algorithm. To overcome the limitation, Sridhar et al. [17] have proposed a partial-update MACE which enables a more efficient algorithm by replacing each iteration agent operator with a faster noniterative update. In partial-update MACE, agents based on regularization priors can be performed using only a single pass of iterative step and a continuously updated initial state.

We apply the idea of partial update to our proposed method and provide detailed solution to each regularization-based agent. Sridhar et al. [17] presented the single iteration of the iterative coordinate descent (ICD) optimization method [41] to efficiently solve MACE equations. In this article, we will explore the use of a single iteration of ISTA for calculating regularization-based agents. Let $\tilde{F}_i(\boldsymbol{v}_i, \boldsymbol{z}_i)$ denote the partial update of agent $F_i(\boldsymbol{v}_i)$, where $\boldsymbol{z}_i$ specifies the initial state. Then come to the single-pass iteration of agent F_1 (12)

$$\begin{aligned} \tilde{F}_1(\boldsymbol{v}_1, \boldsymbol{z}_1) &= H_\lambda\big[\boldsymbol{z}_1 + \mu\boldsymbol{\Phi}^H(\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{z}_1) + \mu'(\boldsymbol{v}_1 - \boldsymbol{z}_1)\big] \\ &\cong H_\lambda\big[\boldsymbol{z}_1 + \mu\mathcal{I}(\boldsymbol{y} - \mathcal{G}(\boldsymbol{z}_1)) + \mu'(\boldsymbol{v}_1 - \boldsymbol{z}_1)\big] \end{aligned} \tag{33}$$

where μ denotes the step-size coefficient of ISTA and $\mu' = \mu\eta$, and $H_\lambda(\cdot)$ denotes the sparse group threshold function, which can be derived from [42]. The regularization parameter λ is empirically updated by the $(K+1)$th largest element of $(\sum_{j=-1,0,1} |\boldsymbol{x}_{j,\sim}|^2)^{1/2}$, where K is the sparsity coefficient of the scene

$$\lambda = \max\left\{\sqrt{\sum_{j=-1,0,1} |\boldsymbol{x}_{j,\sim}|^2}\right\}_{(K+1)} \tag{34}$$

where $\boldsymbol{x}_{j,\sim}$ means the elementwise operate of vector $\boldsymbol{x}_j$. Let $\boldsymbol{z}_i^{k+1}$ be the initial state of agent $\tilde{F}_i$ in the $(k+1)$th iteration, then $\boldsymbol{z}_i^{k+1}$ is decided by the output of $\tilde{F}_i$ in the kth iteration

$$\boldsymbol{z}_i^{k+1} = \tilde{F}_i\big(\boldsymbol{v}_i^k, \boldsymbol{z}_i^k\big). \tag{35}$$

Similarly, the single-pass ISTA iteration of agent F_2' can be formulated as

$$\tilde{F}_2'(\boldsymbol{v}_2, \boldsymbol{z}_2) = \text{threshold}_{\text{MC}}[\boldsymbol{z}_2 + \mu(\boldsymbol{v}_2 - \boldsymbol{z}_2)] \tag{36}$$

where $\text{threshold}_{\text{MC}}(\cdot)$ is the soft threshold function corresponding to MC penalty, which is given by

$$\text{threshold}_{\text{MC}}(\boldsymbol{x}) = \begin{cases} 0, & \eta|x_i| < 1 \\ \theta\dfrac{\eta|x_i| - 1}{\theta - 1}, & 1 \le \eta|x_i| < \theta \\ x_i, & \eta|x_i| \ge \theta. \end{cases} \tag{37}$$

We simply take $\theta = 2$ in this article. In the same way, F_2'' based on dictionary prior and F_3'' based on TV prior can be calculated by single-pass iteration agents $\tilde{F}_2''(\boldsymbol{v}_2, \boldsymbol{z}_2)$ and $\tilde{F}_3''(\boldsymbol{v}_3, \boldsymbol{z}_3)$, respectively.

The partial-update MACE shown in Algorithm 2 can significantly reduce the computational complexity and time of the overall model to a level acceptable for SAR processing. Sridhar et al. [17] had theoretically demonstrated that the partial-update MACE algorithm permits a faster implementation without affecting convergence.

Algorithm 2 Partial-Update MACE-Based SAR Image Reconstruction With Azimuth Ambiguity Suppression Prior, Feature Enhancement Priors, and Despeckling Priors

Input: $\boldsymbol{\Phi}$, $\boldsymbol{y}$, $\boldsymbol{V} = \boldsymbol{V}^0$, $\boldsymbol{u}$, ρ, λ, η, μ

1: $\mu' = \mu\eta$
2: Generate agent F_1.
   $\tilde{F}_1(\boldsymbol{v}_1, \boldsymbol{z}_1) = H_\lambda[\boldsymbol{z}_1 + \mu\mathcal{I}(\boldsymbol{y} - \mathcal{G}(\boldsymbol{z}_1)) + \mu'(\boldsymbol{v}_1 - \boldsymbol{z}_1)]$
3: Generate agent F_2.
   $\tilde{F}_2'(\boldsymbol{v}_2, \boldsymbol{z}_2) = threshold_{MC}[\boldsymbol{z}_2 + \mu(\boldsymbol{v}_2 - \boldsymbol{z}_2)]$ or
   $\tilde{F}_2''(\boldsymbol{v}_2, \boldsymbol{z}_2) = \boldsymbol{\Theta}\boldsymbol{D} \cdot threshold_{\ell_p}[\boldsymbol{\alpha}_2 + \mu(\boldsymbol{v}_2 - \boldsymbol{\Theta}\boldsymbol{D}\boldsymbol{\alpha}_2)]$
   $(\boldsymbol{z}_2 = \boldsymbol{\Theta}\boldsymbol{D}\boldsymbol{\alpha}_2)$
4: Generate agent F_3.
   $F_3' = \text{BM3D}(\boldsymbol{v}_3)$ or
   $\tilde{F}_3''(\boldsymbol{v}_3, \boldsymbol{z}_3) = threshold_{TV}[\boldsymbol{z}_3 + \mu(\boldsymbol{v}_3 - \boldsymbol{z}_3)]$
5: $\boldsymbol{Z} = [\boldsymbol{z}_1\ \boldsymbol{z}_2\ \boldsymbol{z}_3] = \boldsymbol{0}$
6: **while** not converged **do**
7: $\quad \boldsymbol{Z} = \tilde{\boldsymbol{F}}(\boldsymbol{V}, \boldsymbol{Z})$
8: $\quad \boldsymbol{U} = \boldsymbol{G}_{\boldsymbol{u}}(2\boldsymbol{Z} - \boldsymbol{V})$
9: $\quad \boldsymbol{V} = \boldsymbol{V} + 2\rho(\boldsymbol{U} - \boldsymbol{Z})$
10: **end while**

Output: $\hat{\boldsymbol{x}} = \bar{\boldsymbol{v}}$

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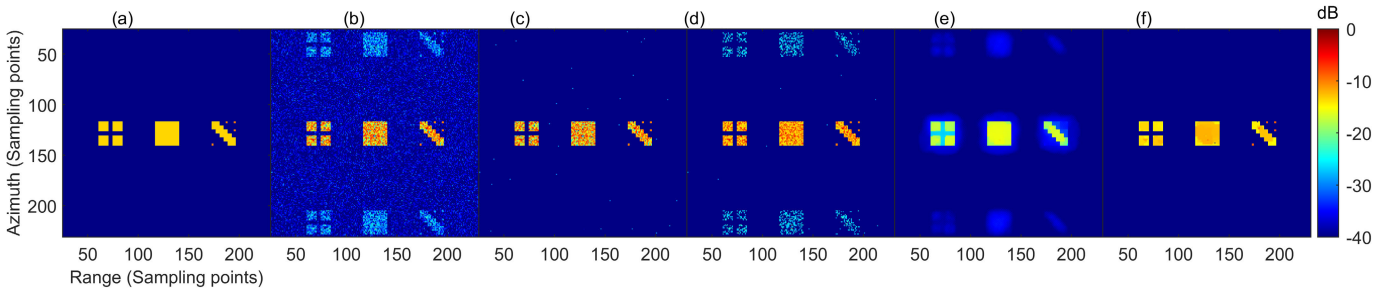


Fig. 3. Reconstruction results of different algorithms and agents. (a) Real value of the scene. (b) Chirp-scaling algorithm. (c) Azimuth ambiguity suppression agent F_1: $\ell_{2,1}$-norm. (d) Feature enhancement agent F_2'': dictionary-based regularization. (e) Despeckling agent F_3'': TV-norm. (f) Proposed MACE-based method: $F_1 + F_2'' + F_3''$.

D. Analysis of Computational Complexity

Know that there are M points in the vector form of raw echo data and N points in the vector form of imaging scene. Suppose no downsampling in azimuth direction, which means $N = M$. As shown in Table I, the computational complexity of the matched filtering method can be expressed as $O(N \log_2 N)$, while the computational complexity of the measurement matrix-based sparse SAR imaging method will reach $O(I_1 N^2 + I_1 N)$, where I_1 denotes the number of iterations for ISTA. However, when using the azimuth-range decouple operator instead of measurement matrix, the computational complexity decreases to $O(I_1 N \log_2 N + I_1 N)$, potentially reducing the overall processing time. Therefore, the azimuth-range decouple operator can speedup the algorithm by an approximate ratio of $r_1 = O(N/\log_2 N)$.

Let I_2 be the number of iterations for calculating individual agents based on regularization priors, the computational complexity attached with calculating agent F_1 can be expressed as $O(I_2 N \log_2 N + I_2 N)$. Suppose the number of iterations for MACE remains to be I_1, and the computational complexity of traditional MACE can be approximately expressed as $O(I_1 I_2 N \log_2 N + I_1 L I_2 N + I_{1\text{LN}})$, where L is the number of agents. The computational complexity of TV or MC regularization used in the mentioned algorithms can be calculated as $O(I_2 N)$, which is not negligible as iterations I_2 can be hundreds of times. In the case of partial-update MACE, no more iterations are required for individual agents, so that the computational complexity will be $O(I_1 N \log_2 N + 2I_{1\text{LN}})$. Therefore, the partial-update idea can speedup the algorithm by an approximate ratio of $r_2 = O(I_2)$. Consequently, the partial-update MACE SAR reconstruction algorithm has more operations than sparse SAR imaging method. However, the proposed algorithms can adapt to the variant priors and can realize the integrated feature enhancement, denoising, and imaging reconstruction, hence the additional computation burden is acceptable.

TABLE I
COMPUTATIONAL COMPLEXITY OF DIFFERENT ALGORITHMS

Algorithms	Computational Complexity
Matched filtering	$O(N \log_2 N)$
Sparse imaging($\mathbf{\Phi}, \mathbf{\Phi}^H$)	$O(I_1 N^2 + I_1 N)$
Sparse imaging($\mathcal{G}, \mathcal{I}$)	$O(I_1 N \log_2 N + I_1 N)$
MACE	$O(I_1(I_2 N \log_2 N + L I_2 N + L N))$
partial-update MACE	$O(I_1(N \log_2 N + 2L N))$

V. EXPERIMENTS AND ANALYSIS

The performance of the proposed method is examined through experiments on both simulated and QILU-1 satellite real data. Simulation experiments are conducted to test the effectiveness of the proposed MACE-based SAR reconstruction method that integrates multiple prior information for improved image quality. Furthermore, real-data experiments are conducted to assess the performance of the proposed method compared to other SAR imaging algorithms, highlighting the superiority of our method.

A. Simulation

To demonstrate the feasibility of MACE mechanism, we conduct simulation experiments by operating each agent independently and then integrate all the agents using the proposed method. To highlight the individual contributions of each agent, we set the reconstruction result of the chirp scaling algorithm (CSA) as the input for each agent. Comparative analysis of both visual and quantitative results verifies the effectiveness of the proposed MACE method in enhancing SAR imaging quality.

First, we generate a 256×256 pixels scene with distributed targets and point targets, as shown in Fig. 3(a). An SNR=15 dB Gaussian white noise is added to the raw data, according to (1). Azimuth ambiguity, noise, and speckle are displayed in the reconstruction result of CSA, as shown in Fig. 3(b). Besides the reconstruction and azimuth ambiguity suppression agent F_1, we take dictionary-based agent F_2'' for feature enhancement and set a shape-based (SB) dictionary [28] to preserve the edge of targets. Agent F_3'' based on TV-norm is chosen for despeckling since it is more efficient in calculating small scene than BM3D. In Fig. 2(b), the agents $(F_1 + F_2'' + F_3'')$ connected by dashed lines are used for simulation experiments. The parameters in the proposed method are set to $\eta = 1$, $\rho = 0.1$, and $\boldsymbol{u} = [1, 1, 1]$. The maximum iteration times $I_1 = 100$.

The operating results of azimuth ambiguity suppression agent F_1, feature enhancement agent F_2, and despeckling agent F_3 are shown in Fig. 3(c)–(e), respectively. F_1 is an SAR image reconstruction agent with $\ell_{2,1}$ prior to suppress azimuth ambiguity. According to Fig. 3(c), the ambiguous areas are apparently repressed under the effects of agent F_1. F_2 is a feature enhancement agent generated from shape-based

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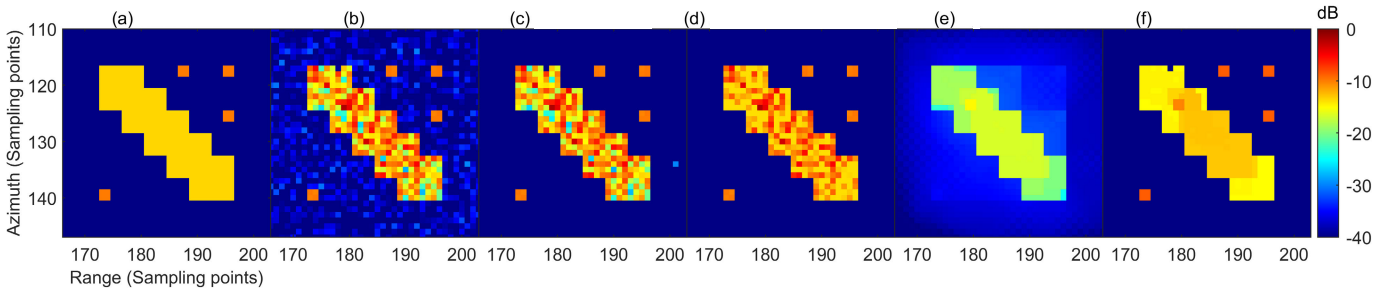


Fig. 4. Partial enlargement of the reconstruction results in Fig. 3. (a) Real value of the scene. (b) CSA. (c) Azimuth ambiguity suppression agent F_1: $\ell_{2,1}$-norm. (d) Feature enhancement agent F_2'': dictionary-based regularization. (e) Despeckling agent F_3'': TV-norm. (f) Proposed MACE-based method: $F_1 + F_2'' + F_3''$.

TABLE II
EVALUATION OF SIMULATION RECONSTRUCTION RESULTS

Algorithms	AASR(dB)	TBR(dB)	EPI	ENL
CSA	-18.9	23.5	1.15	0.96
F_1: $\ell_{2,1}$ [5]	-39.9	38.1	1.19	0.96
F_2: dictionary [28]	-20.3	43.3	**1.36**	1.27
F_3: TV	-20.1	13.8	0.46	**31.4**
proposed MACE	**-55.2**	**56.3**	1.27	27.1

dictionary as prior. Fig. 3(d) demonstrates that F_2 directly improves TBR and the shape of targets. F_3, which is a despeckling agent, introduces regularization prior TV-norm. The speckle is successfully eliminated and the region-based features are enhanced through agent F_3 as shown in Fig. 3(e). Fig. 3(f) shows the reconstruction result of combining all three agents by the proposed method, which achieves the consensus equilibrium among (c)–(e).

In Fig. 4, the partial enlargement of imaging results is presented, allowing the visual effect of each prior to be observed clearly. Despite the positive impact of each agent, we cannot ignore their potential drawbacks. In Figs. 3(c) and 4(c), the azimuth ambiguity suppression agent leaves holes in the distributive targets because of the oversparsity of the scene. In Fig. 4(d), the SB dictionary prior enhances the shape-based features of targets but intensifies the ambiguity areas as well. In Fig. 4(e), background and scatters are excessively smoothed, resulting in the loss of point targets. Nevertheless, the proposed MACE method presented in Fig. 4(f) strikes a balance among the three agents, capitalizing on their strengths while mitigating their weaknesses. Consequently, the synergistic effects of multiple agents through MACE mechanism could lead to the improvement in imaging quality while simultaneously overcoming the limitations of individual agents.

Indicators are introduced for objective evaluation. In order to quantitatively evaluate the azimuth ambiguities suppression ability of the $\ell_{2,1}$-norm prior, we select the azimuth ambiguity-to-signal ratio (AASR) [5] as the evaluation indicator in this article, and the AASR value of the ith ambiguous area is defined as follows:

$$\mathrm{AASR} = 10\lg\left(\frac{1/N_{\mathcal{M}}\sum_{(i,j)\in\mathcal{M}} I_{i,j}}{1/N_{\mathcal{A}}\sum_{(i,j)\in\mathcal{A}} I_{i,j}}\right) \tag{38}$$

where $\boldsymbol{I} = \{I_{i,j}\} = \{|\boldsymbol{X}|_{i,j}^2\}$ denotes the intensity matrix of the image, and $\boldsymbol{X}$ is the matrix form of image vector $\boldsymbol{x}$. $N_{\mathcal{A}}$ is the numbers of the pixel in the main imaging area $\mathcal{A}$, and $N_{\mathcal{M}}$ is the numbers of the pixel in the ith ambiguous area $\mathcal{M}_j$. AASR is expected to be low for good ambiguity suppression effect.

To further analyze the feature enhancement performance of dictionary-based prior, the TBR [43] and edge preservation index (EPI) [44] are used to evaluate the additive noise suppression effect and texture preserving effect, respectively

$$\mathrm{TBR} = 10\lg\left(\frac{I_{\mathrm{target}}}{I_{\mathrm{background}}}\right) \tag{39}$$

$$\mathrm{EPI} = \frac{\sum_{l=1}^{L}|\hat{\boldsymbol{x}}|_{l,1} - |\hat{\boldsymbol{x}}|_{l,2}}{\sum_{l=1}^{L}|\boldsymbol{x}|_{l,1} - |\boldsymbol{x}|_{l,2}}. \tag{40}$$

I_{target} and $I_{\mathrm{background}}$ denote the total intensity of the evaluating target and clutter area, respectively, $|\hat{\boldsymbol{x}}|_{l,1}$ and $|\hat{\boldsymbol{x}}|_{l,2}$ denote the amplitude of the reconstructed image on both sides of the edge, and $|\boldsymbol{x}|_{l,1}$ and $|\boldsymbol{x}|_{l,2}$ denote the amplitude of the reference image on both sides of the edge. Let the original scene [Fig. 3(a)] be the reference image in simulation experiments. A higher TBR value indicates a lower background noise. The larger the EPI value than 1, the better the edge preserve effect.

Equivalent number of looks (ENL) [45] is commonly used to measure the speckle suppression of SAR imaging methods

$$\mathrm{ENL} = \frac{\mu^2(\boldsymbol{I})}{\sigma^2(\boldsymbol{I})}. \tag{41}$$

$\mu(\cdot)$ calculates the mean and $\sigma(\cdot)$ calculates the standard deviation. When the ENL value is bigger, it indicates the image is smoothed well.

The results are listed in Table II. The proposed method significantly outperforms others in terms of AASR and TBR, while it does yield slightly lower ENL value compared to agent F_3'' of TV prior. Nevertheless, the proposed method maintains an EPI value higher than 1, as F_3'' gets merely 0.46 for EPI. Since F_2'' uses the shape-based dictionary as prior to enhance edges of the targets, F_2'' gets the highest EPI value, followed by our proposed method. The trade-off between EPI and ENL is made by MACE mechanism. Thus, the proposed MACE-based method can simultaneously achieve low ambiguity, high TBR, and low-coherence speckle SAR imaging, noted as the ability of integrating multiple agents to achieve better reconstruction quality.

Experimental evidence verifying the rationality of convergence analysis in Section IV-B is provided as follows. Fig. 5(a) convergence curve illustrates the relationship between the change in the output result and the number of iterations

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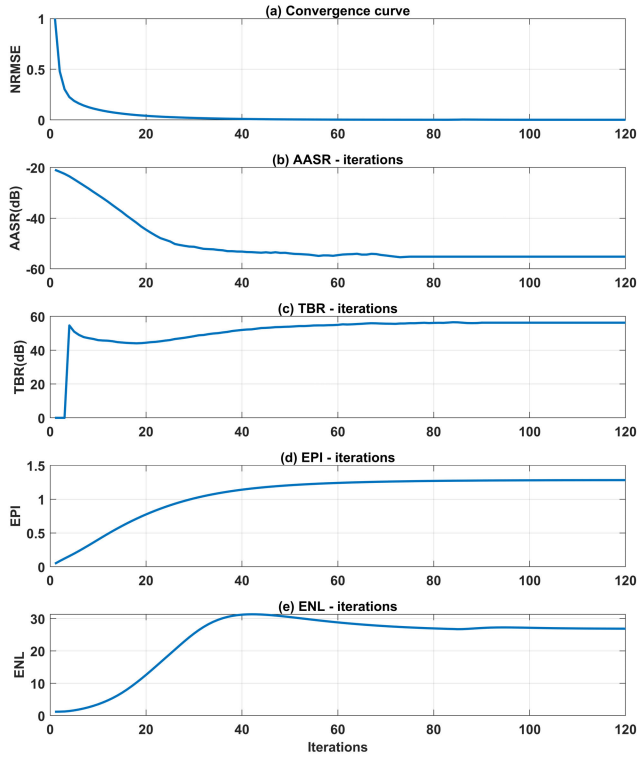


Fig. 5. Convergence analysis of the proposed method based on simulated data. (a) Convergence curve. (b) AASR iterations. (c) TBR iterations. (d) EPI iterations. (e) ENL iterations.

during the iterative process. The change made by each iteration is measured by the normalized root mean squared error (NRMSE) of iteration input and output

$$\text{NRMSE}(k) = \frac{\|\boldsymbol{V}^k - \boldsymbol{V}^{k-1}\|_2}{\|\boldsymbol{V}^{k-1}\|_2} \tag{42}$$

where $\boldsymbol{V}^{k-1}$, $\boldsymbol{V}^k$ denote the output results of the $(k-1)$th and kth iteration respectively, and $\|\cdot\|_2$ denotes the ℓ_2-norm of the vector. As the number of iterations increases, the RNMSE dramatically diminishes and approaches a near-zero value. The significance of this trend is that it confirms that the output of each subsequent iteration fluctuates around the fixed point of Mann iteration. Fig. 5(b)–(e) plots the trend of evaluation indicators (AASR, TBR, EPI, and ENL) with iterations, respectively. The TBR values of the initial three iterations are set to zero to bypass any nonreal values since the weak targets are buried within the noise and are difficult to detect. As the energy accumulates, the intensity of the targets after the fourth iteration is sufficient to calculate the TBR. It is clear that the values of such indicators remain relatively consistent after 100× of iterations, suggesting a stable imaging quality after the maximum iteration times set in simulation experiments. Consequently, with each iteration of the process, the outputs progressively approach the anticipated optimal imaging result, thereby ensuring the proposed algorithm's capability to converge.

B. Real Data of QILU-1 Satellite

To illustrate the performance of the proposed method in ambiguity suppression, we use QILU-1 real SAR data [31] at certain beam position. The reconstruction result of CSA is in Fig. 6(a). Highly intense targets cause azimuth ambiguities, which are explicitly shown in areas A and C. Simultaneously, the reconstruction quality is severely compromised by excessive noise and speckle. Area B contains a wide range of distributed targets, and the imaging quality is highly affected by speckles. Agents $F_1 + F_2' + F_3'$, which are connected in Fig. 2(b) by solid lines, are introduced for real-data experiments. Three priors, including $\ell_{2,1}$-norm (12), MC penalty (14), and BM3D filter (17), are integrated by the proposed method for real data. The parameters in the proposed method are set to $\eta = 1$, $\rho = 0.4$ for the first 30 iterations and $\rho = 0.05$ for the next 20 iterations (to improve the convergence speed), $\boldsymbol{u} = [1, 1, 0.5]$. The reconstruction result of the proposed method is presented in Fig. 6(b). The effectiveness of the proposed MACE-based method in azimuth ambiguity suppression, enhancing point-based features and despeckling, is supported.

Fig. 7 plots the amplitudes of reconstruction results at certain azimuth position, which is marked by dashed line in Fig. 6(a) and (b). The performance of azimuth ambiguity suppression, feature enhancement, denoising, and despeckling can be directly illustrated by comparison. For azimuth ambiguity suppression, the proposed method effectively eliminates the ambiguity area in the scene. For region-based features enhancement and despeckling, the CSA method produces imaging results affected by speckles, which can disrupt the smoothness of the distributed target. In contrast, our proposed method ensures the consistency of reconstructed amplitudes, resulting in more satisfying imaging results. For point-based feature enhancement, our innovative method ensures the preservation of the reconstructed amplitude of the strong scattering targets. In addition, the SAR reconstructed method we proposed has an excellent noise reduction capability, demonstrating a great improvement in TBR and delivering superior visual performance.

Then, we reconstruct the scene with the CSA, ℓ_1-norm regularization [1], $\ell_{2,1} - \ell_1$ composite regularization [5], MC-TV composite regularization [6], as well as PnP-BM3D [10] for comparative experiments. Three areas (A, B, and C), highlighted in the original scene in Fig. 6(a), provide examples for demonstrating the imaging performance of each SAR reconstruction method. The experimental results are displayed in Fig. 8. The first row presents partial enlargement images of area A. The azimuth ambiguity of intense targets is suppressed in Fig. 8(c) and (f), which suggests that the proposed method effectively integrates the function of $\ell_{2,1}$-norm prior. The second row presents distributed targets in area B. The proposed MACE method increases the smoothness of region targets in Fig. 8(l). Although using BM3D as PnP prior independently leads to an improved performance in despeckling, it also results in excessive smoothing of noise in Fig. 8(k). The third row presents area C. The point-based features are significantly enhanced and noise is sufficiently suppressed in Fig. 8(r). Compared with the composite regularization method MC-TV, the proposed method achieves better despeckling performance by introducing BM3D filters as prior. Compared with the composite regularization method $\ell_{2,1} - \ell_1$, the proposed method

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(a)



(b)

Fig. 6. Reconstruction result of QILU-1 satellite real data. (a) CSA. (b) Proposed MACE-based method: $F_1 + F_2' + F_3'$.

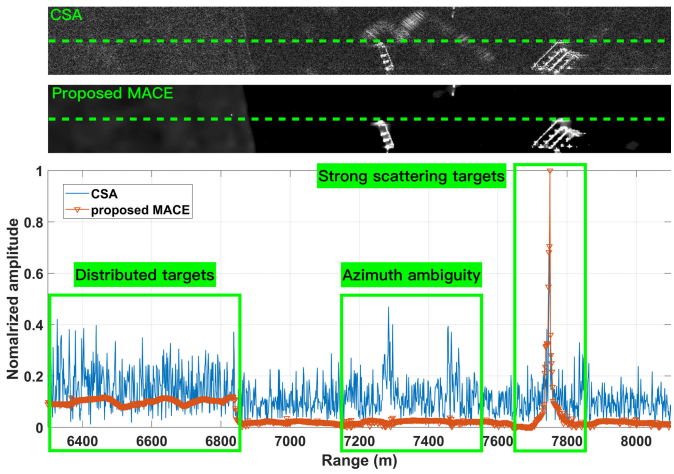


Fig. 7. Quantitative analysis of dashed line area in Fig. 6.

effectively enhances point-based features of strong scattering targets by utilizing MC penalty as prior and enhances region-based features via despeckling prior BM3D filter.

Fig. 9 presents a comparison of the reconstructed amplitudes of different SAR imaging algorithms. Specifically, Fig. 9(a)–(c) corresponds to the positions, marked by dashed line in area A–C, respectively, as shown in Fig. 8(a), (g), and (m). The figures help visualize different notable targets or areas in the imaging scene. Fig. 9(a) plots an ambiguity area on the left and urban targets, (b) plots the distributive targets, and (c) plots the highly intense targets on the water. Compared with ℓ_1-norm regularization method, the proposed method eliminates the bias effect of reconstructed amplitudes [6], ensuring that the amplitudes of highly intense targets are similar to CSA. Furthermore, our method integrates ambiguity suppression and despeckling functions that further enhances imaging quality. Compared with the $\ell_{2,1}-\ell_1$ regularization, the proposed method also performs better in the accuracy of reconstructed amplitudes. Additionally, it enhances the smoothness of distributive targets. Compared with MC-TV regularization, our method outperforms its despeckling function in maintaining the consistency of reconstructed amplitudes across distributed targets. Moreover, the azimuth ambiguity suppression function also provides a significant advantage that differentiates the proposed method from MC-TV regularization. Compared with PnP-BM3D method, the proposed method exhibits superior performance in preserving texture

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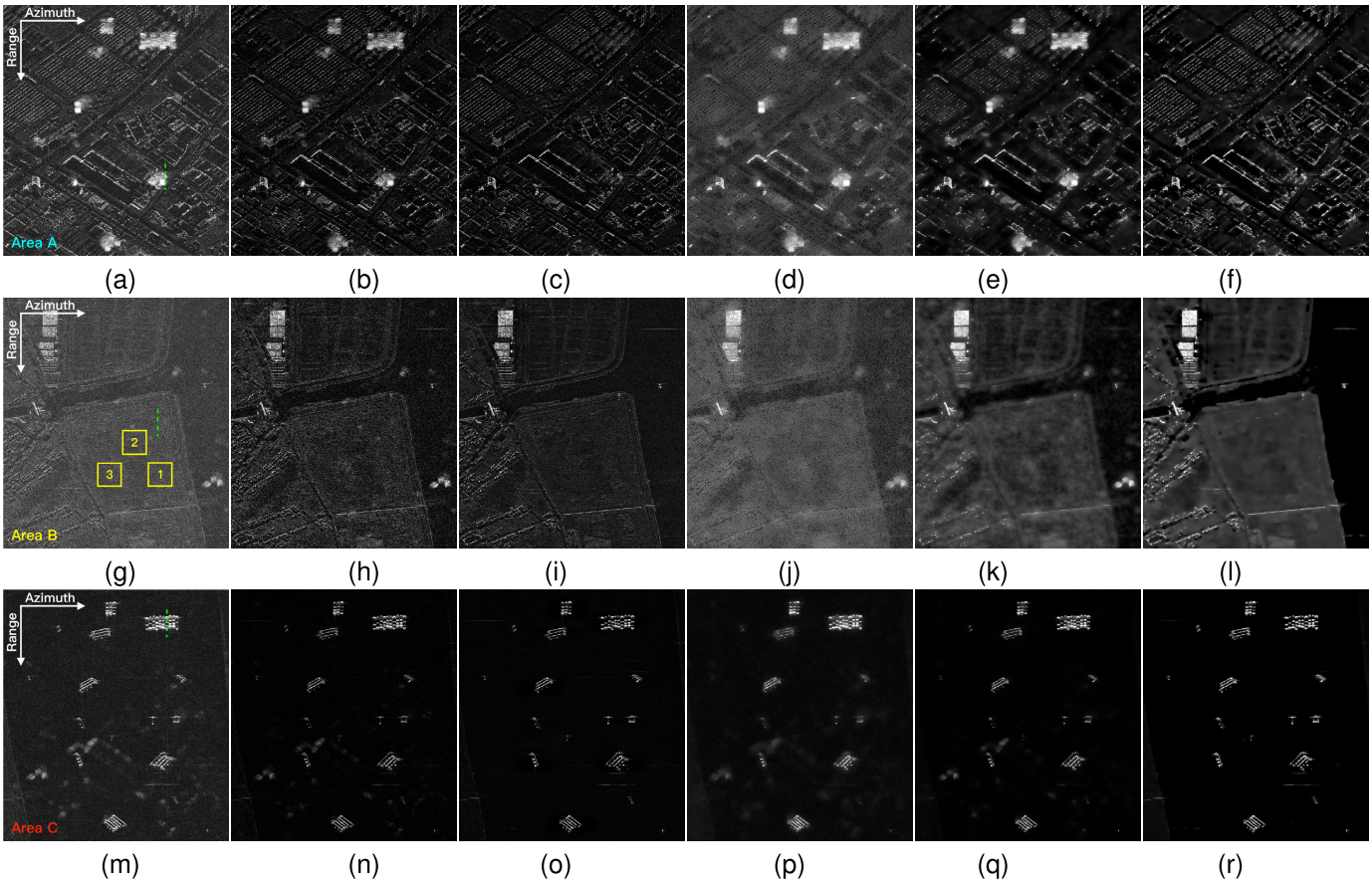


Fig. 8. QILU-1 real-data reconstruction results of different algorithms and agents. First row: area A, second row: area B, third row: area C. (a), (g), and (m) Chirp-scaling algorithm, (b), (h), and (n) ℓ_1-norm regularization [1], (c), (i), and (o) $\ell_{2,1} - \ell_1$ composite regularization [5], (d), (j), and (p) MC-TV composite regularization [6], (e), (k), and (q) PnP-BM3D [10], (f), (l), and (r) proposed MACE-based reconstruction method.

details and in eliminating ambiguity from the reconstructed results. To sum up, the superiority of MACE to improve reconstruction image quality and simultaneously integrating various priors has been corroborated.

Quantitative analysis results are shown in Table III, whereby the enhancement of different types of features is evaluated numerically. AASR is utilized to evaluate the azimuth ambiguity suppression performance. For evaluating point-based features, TBR and entropy of the full image (ENT) [46] are utilized and for region-based features, ENL and mean target edge strength (MTES) [47] are utilized. Fig. 8(g) depicts the locations of distributed regions B-1, B-2, and B-3 with squares. Evaluating the ENL values of B-1, B-2, and B-3 in SAR imagery provides a quantitative measure of the despeckling capabilities of the imaging algorithm. The region-based features are enhanced as speckles are suppressed

$$\mathrm{ENT} = \ln(E_x) - (1/E_x) \sum_{i,j} I_{i,j} \ln\left(I_{i,j}\right) \tag{43}$$

where total energy $E_x = \sum_{i,j} I_{i,j}$

$$\mathrm{MTES} = \sum_{l \in \mathcal{L}} |\hat{\boldsymbol{x}}|_l \tag{44}$$

where $\mathcal{L}$ denotes the set of edge elements with amplitudes above $x_{\min}$ and $|\hat{\boldsymbol{x}}|_l$ denotes the amplitude of the reconstructed image. The Sobel operator is used to generate the edge map for targets. Then the average of all edge magnitudes above a minimum threshold is denoted as the MTES [47].

We would expect TBR to be high and ENT to be low in the point-like component. It would also be expected for ENL and MTES to be high in the region-like components. Results show that our proposed method outperforms other methods in terms of AASR, TBR, ENT, and MTES, indicating better azimuth ambiguity suppression, denoising, speck smoothing, texture retention, and edge preservation of the reconstructed image. Despite the ENL values of PnP-BM3D being significantly high, the proposed method shows a relatively higher ENL value compared to the other methods, demonstrating satisfied speckle smoothing effect. Nevertheless, PnP-BM3D fails to perform well when evaluating point-based features, including TBR and ENT. Consequently, the proposed method outperforms other methods in both point-based and region-based features enhancement, and achieves the best SAR reconstruction quality.

Table III also provides the processing time of each SAR imaging method, implemented on MATLAB R2021a. The configuration of the experimental platform consists of an Intel Core i9-13900K central processing unit (CPU) and an NVIDIA RTX4090 graphics processing unit (GPU). The maximum iteration times I_1 is set to 50 for all the sparse processing model-based methods. It is important to clarify that the measurement of imaging time is based on a sufficient

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TABLE III
EVALUATION OF REAL-DATA RECONSTRUCTION RESULTS
BEST RESULTS ARE SHOWN IN BOLD

Algorithms	AASR(dB)	TBR(dB)	ENT		ENL			MTES	Times(min)
	Area A	Area C	Area A	Area C	Area B-1	Area B-2	Area B-3	Area B	
CSA	-18.7	10.6	10.3	7.16	0.89	0.98	0.98	0.0350	0.70
ℓ_1 regularization [1]	-20.0	17.9	9.06	6.45	0.22	0.21	0.21	0.0243	38
$\ell_{2,1}-\ell_1$ [5]	-33.1	17.3	8.72	6.46	0.43	0.48	0.51	0.0283	50
MC-TV [6]	-19.8	11.3	10.2	6.52	3.32	2.92	2.78	0.0357	58
PnP-BM3D [10]	-19.4	11.0	10.3	6.99	**83.6**	**63.0**	**66.0**	0.0549	61
proposed MACE	**-34.5**	**21.5**	**8.23**	**6.24**	28.3	27.5	22.3	**0.0594**	44

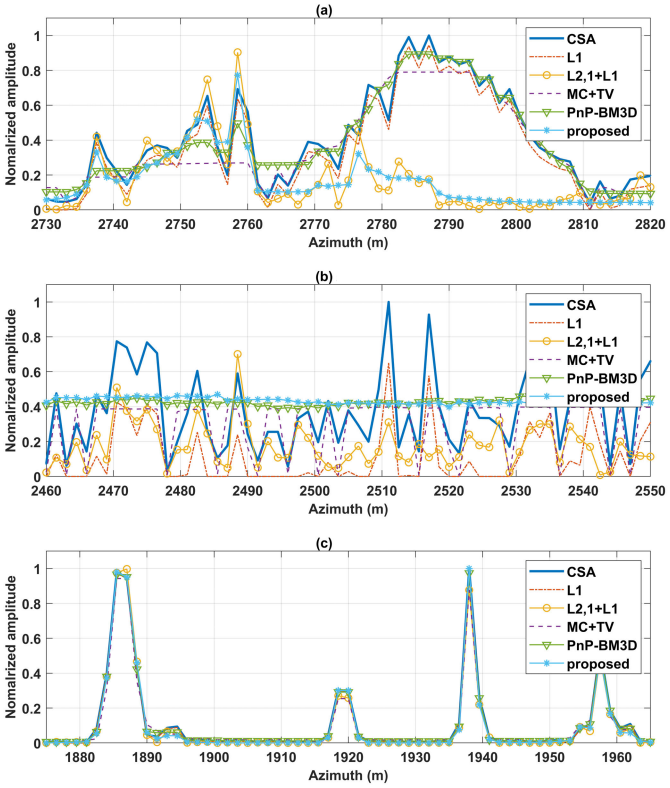


Fig. 9. Quantitative analysis of dashed line area in (a)–(c) areas A–C displayed in Fig. 8.

number of iterations and a personal computer, which has limited computing power. Despite the fact that there are various techniques for accelerating the algorithms, we primarily focus on highlighting the essential features and performance of the proposed method. Therefore, the timing results provided in the table should be primarily used as a reference for analyzing the running time of different algorithms under similar conditions. The imaging time of the MACE method can be further reduced by setting a sequence of ρ [40] or a matrix $\boldsymbol{H}$ instead of ρ [16] to reduce the iteration times, improving the computing power of the imaging system, and using compressive sensing acceleration techniques [48].

Although the imaging times of sparse imaging model-based methods are several times longer than that of the matched filter method, the reconstruction quality is greatly improved. Due to the independent nature of agent computations during each iteration of the proposed MACE algorithm, we are able to realize significant reductions in running time by parallelizing these computations. In this way, while under the sufficient processing resource constraints, the total processing time of the proposed algorithm is dominantly determined by the agent with the highest computational complexity. However, composite regularization methods, such as $\ell_{2,1}-\ell_1$ and MC-TV, which are solved by variable-splitting ADMM algorithm [6], still require computing penalties sequentially. Consequently, the total processing time depends on the sum of all the corresponding priors. Compared with the PnP method, the time-consuming BM3D filter calculation can be executed simultaneously with the image reconstruction and ambiguity suppression agent in the MACE-based method, thus speeding up the overall operation. This explains why the MACE-based method, despite similar computational complexity, operates significantly faster than composite regularization and PnP-BM3D methods.

VI. CONCLUSION

The proposed SAR image reconstruction method based on MACE mechanism integrates functions by unifying agents of both regularization and PnP priors. Mann iteration and partial-update idea are introduced to enhance algorithm efficiency. The convergence and computational complexity of the proposed algorithm are analyzed and demonstrated through experiments. In this article, we combine priors of three distinct function, including the $\ell_{2,1}$-norm for azimuth ambiguity suppression, MC penalty or dictionary-based prior for point-based or shape-based features enhancement respectively, and TV-norm or BM3D filter for despeckling, thereby significantly improving the imaging quality.

Our experimental results on simulated data demonstrate that the MACE approach can solve problems for which there is no corresponding regularized optimization and can in some cases achieve consensus results that are better than any of the individual operators. Experimental results of both simulated and QILU-1 satellite SAR data have validated the exceptional performance of the proposed method.

In this article, we categorize priors into three sets [see Fig. 2(b)] to obtain high-quality SAR images. However, our model can introduce various priors based on different objectives, not limited to just these three categories. When considering the use of different priors, the primary concern is the algorithmic convergence, which has been extensively discussed in this article. In future research, we will explore integrating more priors, such as deep learning networks or complex-valued optimizations, into the MACE-based SAR

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imaging method to further enhance the algorithm's performance and demonstrate its scalability.

APPENDIX

Proof of Theorem 3. Let $e_1, \ldots, e_{NL}$ denote the eigenvalues of the Jacobian matrix of $\boldsymbol{T}$ at $\boldsymbol{W}$. Since the real part of e_i is less than 1, there exists $0 < c < 1$ such that $e_i = a_i + jb_i$ and $a_i \leq 1 - c < 1$. j denotes the imagery unit and $a_i, b_i \in \mathbb{R}$

$$\begin{aligned} |(1-\rho) + \rho e_i|^2 &= [(1-\rho) + \rho a_i]^2 + \rho^2 b_i^2 \\ &\leq [(1-\rho) + \rho(1-c)]^2 + \rho^2 b_i^2 \\ &= 1 + \rho^2\left(b_i^2 + c^2\right) - 2\rho c. \end{aligned} \tag{45}$$

Take $\rho < 2c/(b_i^2 + c^2)$, then $g_i = (1-\rho) + \rho e_i$ lies strictly inside the unit disk, which means there exists $r < 1$ that controls the eigenvalues during iterations $|g_i| \leq r < 1$.

Let $\boldsymbol{A}$ be the Jacobian matrix of $\boldsymbol{T}$ at the fixed point $\boldsymbol{W}$, which means $\boldsymbol{T}(\boldsymbol{V}) = \boldsymbol{T}(\boldsymbol{W}) + \boldsymbol{A}(\boldsymbol{V} - \boldsymbol{W}) + o(\boldsymbol{V} - \boldsymbol{W})$. Equation (32) turns Mann iteration to a linear map $\boldsymbol{V}^{k+1} - \boldsymbol{W} = [(1-\rho)\boldsymbol{I} + \rho\boldsymbol{A}](\boldsymbol{V}^k - \boldsymbol{W})$. Let $\boldsymbol{A}_\rho = (1-\rho)\boldsymbol{I} + \rho\boldsymbol{A}$, then the eigenvalues of $\boldsymbol{A}_\rho$ can be written as $g_i = (1-\rho) + \rho e_i$. The Schur triangulation gives a unitary matrix $\boldsymbol{Q}$ and an upper triangular matrix $\boldsymbol{U}$ with $\boldsymbol{A}_\rho = \boldsymbol{Q}^H \boldsymbol{U} \boldsymbol{Q}$. The diagonal entries of $\boldsymbol{U}$ should be $g_1, g_2, \ldots, g_{NL}$. Let $\boldsymbol{G} = \operatorname{diag}\{g_i\}$ and $\boldsymbol{U} = \boldsymbol{G} + \boldsymbol{U}'$, then the compatibility of matrix and vector norm gives

$$\begin{aligned} |\boldsymbol{A}_\rho(\boldsymbol{V} - \boldsymbol{W})| &= |\boldsymbol{Q}^H\left(\boldsymbol{G} + \boldsymbol{U}'\right)\boldsymbol{Q}(\boldsymbol{V} - \boldsymbol{W})| \\ &= |\boldsymbol{G}(\boldsymbol{V} - \boldsymbol{W}) + \boldsymbol{Q}^H\boldsymbol{U}'\boldsymbol{Q}(\boldsymbol{V} - \boldsymbol{W})| \\ &\leq \|\boldsymbol{G}\|_2|\boldsymbol{V} - \boldsymbol{W}| + \|\boldsymbol{Q}^H\boldsymbol{U}'\boldsymbol{Q}\|_2|\boldsymbol{V} - \boldsymbol{W}| \\ &\leq \left(r + \|\boldsymbol{Q}^H\boldsymbol{U}'\boldsymbol{Q}\|_2\right)|\boldsymbol{V} - \boldsymbol{W}| \end{aligned} \tag{46}$$

where $|\cdot|$ denotes the magnitude of the vector and $\|\cdot\|_2$ denotes the ℓ_2-norm of the matrix. Suppose $\boldsymbol{F} = \operatorname{diag}\{f_{ii}\}$ is a $NL \times NL$ diagonal matrix and vector $\boldsymbol{x}^0$ lies in the neighborhood $\mathcal{B}$ of the origin. By setting $\boldsymbol{V}^0 - \boldsymbol{W} = \boldsymbol{Q}^H\boldsymbol{F}^H\boldsymbol{x}^0$, as in (46), we obtain

$$\begin{aligned} |\boldsymbol{F}\boldsymbol{Q} \cdot \boldsymbol{A}_\rho\left(\boldsymbol{V}^0 - \boldsymbol{W}\right)| &= |\boldsymbol{F}\boldsymbol{Q}\boldsymbol{A}_\rho\boldsymbol{Q}^H\boldsymbol{F}^H\boldsymbol{x}^0| \\ &\leq \left(r + \|\boldsymbol{F}\boldsymbol{U}'\boldsymbol{F}^H\|_2\right)|\boldsymbol{x}^0|. \end{aligned} \tag{47}$$

Since $\{\boldsymbol{F}\boldsymbol{U}'\boldsymbol{F}^H\}_{pq} = u_{pq}f_{pp}f_{qq}$, where u_{pq} is the entry of $\boldsymbol{U}'$ at the pth row and qth column, we can constrain the norm $\|\boldsymbol{F}\boldsymbol{U}'\boldsymbol{F}^H\|_2$ by letting

$$f_{ii} = \left[\frac{1-r}{2u_{\max}\mathrm{NL}}\right]^i \tag{48}$$

where $u_{\max}$ denotes the maximum magnitudes of entries of $\boldsymbol{U}'$. Then we can get

$$\begin{aligned} |\boldsymbol{F}\boldsymbol{Q}\boldsymbol{A}_\rho\boldsymbol{Q}^H\boldsymbol{F}^H\boldsymbol{x}^0| &\leq \left(r + \|\boldsymbol{F}\boldsymbol{U}'\boldsymbol{F}^H\|_2\right)|\boldsymbol{x}^0| \\ &\leq \left(r + \mathrm{NLu}_{\max} \cdot \frac{1-r}{2u_{\max}\mathrm{NL}}\right) \\ &= \beta|\boldsymbol{x}^0| \end{aligned} \tag{49}$$

where $\beta < 1$ is a constant. Thus,

$$|\boldsymbol{F}\boldsymbol{Q} \cdot \boldsymbol{A}_\rho^k\left(\boldsymbol{V}^0 - \boldsymbol{W}\right)| \leq \beta^k|\boldsymbol{x}^0|. \tag{50}$$

In other words, $\boldsymbol{F}\boldsymbol{Q}\boldsymbol{A}^k\boldsymbol{Q}^H\boldsymbol{F}^H\boldsymbol{x}^0$ shrinks to the origin when iteration number k increases, which means $\boldsymbol{A}_\rho^k\boldsymbol{Q}^H\boldsymbol{F}^H\boldsymbol{x}^0$ converges to the origin. Let $\boldsymbol{V}^0 = \boldsymbol{W} + \boldsymbol{Q}^H\boldsymbol{F}^H\boldsymbol{x}^0$, then

$$\boldsymbol{V}^k - \boldsymbol{W} = \boldsymbol{A}_\rho^k\left(\boldsymbol{V}^0 - \boldsymbol{W}\right) = 0|_{k\to\infty}. \tag{51}$$

Thus, $\boldsymbol{V}^k$ converges to $\boldsymbol{W}$ for all $\boldsymbol{V}^0$ in the neighborhood $\boldsymbol{Q}^H\boldsymbol{F}^H\mathcal{B}$ of $\boldsymbol{W}$.

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